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Causal structure-enhanced branch neural networks for interpretable and robust regression

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ABSTRACT

This paper introduces causal information into neural networks for training and inference to improve the performance of models for stable prediction. Existing neural networks only consider correlation prediction and overlook causal relationships between variables, resulting in spurious correlations between non-causal features and target variables. To solve this problem, we propose a novel framework of **B**ranch **N**eural **N**etworks Based on **C**ausal **C**onstraints (**BNN-CC**), which contains a data preprocessing module, a causal grouping module, and a branch neural network module. The causal grouping module focuses on discovering potential causal relationships between variables from observational data, thus causally grouping the target's parent and ancestor variables. The features obtained through causal grouping are subsequently used as inputs to branch neural networks, where each branch corresponds to the representation of a specific causal group. Finally, we conduct extensive and diverse experiments. The experimental results demonstrate that BNN-CC outperforms existing baselines on both regression and classification tasks, achieving a coefficient of determination 0.913 and an accuracy of 0.916 on the IHDP and Twins datasets, respectively. The source code is publicly available at https://github.com/cjqshu/BNNCC-codes.

1. Introduction

Neural networks, as nonlinear statistical modeling tools, have been widely applied in fields such as computer vision, natural language processing (LeCun et al., 2015), and causal representation learning (Li et al., 2024; Schölkopf et al., 2021; Wu et al., 2022). In the context of supervised learning, the main goal of a neural network is to learn an optimal set of parameters that minimize the loss between the predicted and actual values of the target (label). In general, the model implicitly assumes a causal graph between features and the target variable, where mutually independent features are considered causes of the target variable, denoted as $x_i (i = 1, \dots, p)$ and Y for features and target, as shown in Fig. 1(a). However, the true causal graph is as illustrated in Fig. 1(b), where not all features are mutually independent (e.g., x_1 is a direct cause of x_3 and x_4), and not all features are actual causes of the target variable. Specifically, there exists a spurious correlation between x_4 and Y due to a backdoor path $x_4 \leftarrow x_1 \rightarrow x_3 \rightarrow Y$ (Pearl et al., 2016). x_4 and Y become independent by conditioning on x_1 or x_3. If all features are used as inputs to the neural network, spurious features that are correlated but not causally related to the target variable may be introduced. This leads to the use of non-causal information for prediction, which we refer to as correlation-based prediction. In general, when the training and testing distributions are not identical (i.e., not independent and identically distributed), correlation-based prediction leads to poor robustness, as the model may perform well on the training set but poorly on the testing set (Kuang et al., 2022).

Accordingly, this work aims to improve the interpretability and robustness of model by integrating information derived from causal constraints. This requires incorporating causal information into both the training and inference processes of neural networks; however, existing work that integrates these two aspects remains limited. In this paper, we propose a novel framework called **B**ranch **N**eural **N**etworks Based on **C**ausal **C**onstraints (**BNN-CC**) to enable causal prediction of the target variable. This paper focuses on how to leverage neural network models to extract and represent causal information, with the main challenge lying in how to effectively train networks under causal constraints. First, we obtain the underlying causal graph from observational data using advanced causal discovery methods. Then, features are causally grouped based on the parent and ancestor sets of the target variable. Finally, a branch neural network is employed to perform feature representation and aggregation for each causal group, with the model further incorporating residual connections and an attention mechanism. To the best

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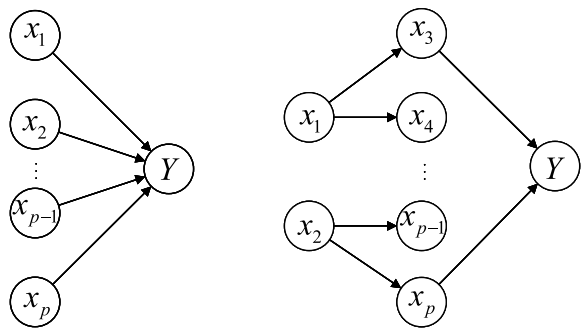


(a) The causal graph under the correlation assumption.

(b) The true causal graph.

Fig. 1. The causal graph under the correlation assumption and the true causal graph.

of our knowledge, we are the first to employ branch neural networks with causal constraints for both regression and classification tasks. Our contributions are summarized as follows:

- This study introduces a causal grouping approach to identify relevant causal structures from the causal graph, which are then modeled and aggregated via branch neural networks. This design improves the interpretability and robustness of both regression and classification tasks.
- We propose a branch neural network based on causal constraints, aiming to perform causal prediction by disentangling spurious correlations among variables. By leveraging the causal grouping of features, it supports the adaptive and dynamic generation of the branch neural network structure. Furthermore, by modifying the prediction heads of the network, the model enables end-to-end learning under causal constraints, effectively accomplishing both regression and classification tasks.
- To validate the effectiveness and generalizability of the proposed model, we conduct extensive and diverse experiments on both synthetic and real-world datasets. The results demonstrate that our model outperforms baseline methods across various regression and classification metrics. Ablation studies confirm the effectiveness of the causal grouping module, OOD generality experiments results show our model's robustness, and sensitivity analysis of hyperparameters identifies the optimal configuration for model performance.

The remainder of this paper is organized as follows. Section 2 reviews the related work. Section 3 presents the notations and formalizes the problem. Section 4 introduces the specific details of our proposed Branch Neural Network Based on Causal Constraints. Section 5 reports the experimental results along with analysis and discussion. Finally, Section 6 concludes the paper.

2. Related work

Causal Discovery from Observational Data. Causal discovery aims to identify causal relationships among variables from observational data. Traditional approaches can be broadly categorized into three classes: constraint-based, functional-based, and score-based methods. Constraint-based causal discovery methods rely on conditional independence tests and identify a class of Markov equivalent Directed Acyclic Graphs (DAGs) (Zhang, 2008). Representative approaches include PC (Peter-Clark) (Spirtes et al., 2000), FCI (Spirtes et al., 2013), and CD-NOD (Huang et al., 2020). Functional-based causal discovery methods distinguish between different directed acyclic graphs (DAGs) within the same Markov equivalence class by imposing additional assumptions on data distributions or functional forms. Representative methods include the Linear Non-Gaussian Acyclic Model (LiNGAM) (Shimizu et al., 2006), Post-Nonlinear (PNL) (Zhang & Hyvarinen, 2012), and Additive Noise Model (ANM) (Hoyer et al., 2008). ANM is suitable for determining causal directions between two variables. When more than two variables are involved, the PC algorithm can first be applied to identify the causal skeleton and partially oriented edges, after which pairwise analysis based on ANM can be used to infer the directions that PC fails to determine. Score-based causal discovery methods evaluate candidate causal graphs using a scoring function based on the observed data, and then search for the graph that achieves the best score. Representative approaches include Greedy Equivalence Search (GES) (Chickering, 2002; Huang et al., 2018) and Dynamic Programming (DP) for optimal graph search (Silander & Myllymaki, 2012). Due to the combinatorial nature of the acyclicity constraint, most score-based methods employ local heuristic strategies to perform the search. Causal discovery methods based on continuous optimization have attracted increasing attention in recent years. Representative approaches include NOTEARS (Zheng et al., 2018), DAG-GNN (Yu et al., 2019), RL-BIC (Zhu et al., 2019), and GOLEM (Ng et al., 2020). NOTEARS is the first method to transform the combinatorial graph search problem into a continuous optimization task. It achieves structure learning through a non-combinatorial optimization via trace exponential augmented lagrangian structure learning. DAG-GNN extends NOTEARS by integrating a neural network function and black-box variational inference, enabling the scoring function to be the Evidence Lower Bound (ELBO). RL-BIC applies a reinforcement learning framework for causal discovery, where an encoder-decoder neural network model serves as the actor to generate candidate DAGs. The generated graphs are evaluated using the Bayesian Information Criterion (BIC), which provides the reward signal to guide the learning process. Gradient-based Optimization of dag-penalized Likelihood for learning linEar dag Models (GOLEM) is a likelihood-based structure learning method with continuous unconstrained optimization.

Studies on Branch Neural Networks. Branch neural networks are a type of neural network architecture that consists of multiple independent branches. These branches process information in parallel and ultimately merge their representations through concatenation or weighted averaging. Such a network architecture improves both the representational power and training efficiency of the model. It has been widely applied in various domains, including image classification (He et al., 2016; Ma et al., 2023; Szegedy et al., 2015; Zhu & Bain, 2017) and disease prediction (Wang et al., 2022; Wang & Yao, 2021). Krizhevsky et al. (2012) introduced AlexNet, which incorporates multiple dual-branch convolutional neural networks to extract and integrate spatial features across different hierarchical levels of the image. This architecture effectively exploits image representations, leading to substantial improvements in classification accuracy. Szegedy et al. (2015) proposed the GoogLeNet architecture based on Network in Network (NiN) (Lin et al., 2013), in which each Inception block consists of four parallel paths. He et al. (2016) proposed Residual Network (ResNet), in which the residual connection can be viewed as two parallel branches. This architecture mitigates gradient vanishing and explosion problems, making the training of deep neural networks feasible. Huang et al. (2017) proposed DenseNet as an extension of ResNet, which extends ResNet by connecting the outputs of different branches through concatenation rather than simple addition. DenseNet facilitates more comprehensive information integration. The above architectures are all based on the design principle of branch neural networks and have demonstrated strong performance across various applications. However, they do not explicitly disentangle high-level semantics. The number of branches in these network models is typically determined empirically or manually, depending on the characteristics of the features and the specific task, such as in the case of multimodal data. Multimodal learning (Ngiam et al., 2011) leverages separate branch networks to process individual modalities (e.g., images, text, audio, or video) and then integrates the information from each branch to accomplish the target task. However, when dealing with

Table 1
Comparison of model characteristics. ✗ and ✓ denote the absence and presence of a characteristic, and "?" denotes unavailable or unclear information.

Model	Parametric Model	Non-Linear	Representation	Causal Information	Regression and Classification
OLS/LR (Best & Wolf, 2013; Craven & Islam, 2011)	✓	✗	✗	?	✗
k-NN (Zhang, 2016)	✗	✓	✗	✗	✓
DT (Ying et al., 2015)	✗	✓	✗	✗	✓
RF (Breiman, 2001)	✗	✓	✗	✗	✓
ABR/ABC (Drucker, 1997; Zhu et al., 2009)	✗	✓	✗	✗	✗
SVR/SVC (Smola & Schölkopf, 2004)	✗	✓	✗	✗	✗
MLP (Kruse et al., 2022)	✓	✓	✓	?	✓
IRMv1 (Arjovsky et al., 2019)	✓	✓	✓	?	✓
NN-SIP (Pros & Vitrià, 2025)	✓	✓	✓	✓	✗
BNN-CC (Ours)	✓	✓	✓	✓	✓

tabular data, it is often challenging to determine the appropriate number of network branches. This necessitates the disentangling of features to adaptively determine the number of branches. To address this problem, our approach leverages causal graphs to disentangle and partition the features.

Our Research Motivation. Despite their strong predictive capabilities, many state-of-the-art neural network models lack interpretability and transparency, which can result in unstable and unreliable performance. To mitigate this problem, we attempt to incorporate interpretable causal information into the network architecture to improve both model transparency and robustness. Our work is related to Neural Networks with Spurious Interactions Prevention (NN-SIP) (Pros & Vitrià, 2025), which proposes a method for estimating causal effects under causally constrained representation learning. This approach not only enhances the interpretability of the model but also improves its ability to perform causal inference tasks, specifically in estimating the causal effect of treatment variable on outcome variable. The main limitation of their model lies in its overly simplistic representation, as it relies solely on fully connected layers. This design may hinder the model's ability to capture informative features effectively and can also lead to problems such as gradient vanishing and explosion during training. Moreover, NN-SIP was not specifically designed for regression and classification tasks. In contrast, our proposed mode BNN-CC differs from NN-SIP in two key aspects. (1) **Tasks**: By integrating causal grouping into representation learning, BNN-CC is capable of effectively handling both regression and classification tasks. (2) **Network architecture**: We incorporate a residual connection to facilitate effective model training and employ an attention mechanism to enable the model to capture important feature information.

To better illustrate the positioning and advantages of our model, we summarize five key characteristics (parametric model, non-Linear, representation, causal Information, and regression and classification) of the proposed approach and compare them with existing models, as shown in Table 1. For a detailed description of the baseline models, please refer to Section 5.1.

3. Notations and problem

This section presents the notations, preliminary concepts, and formal problem definition used throughout this paper.

3.1. Notations

Table 2 lists the notations used in this paper along with their corresponding descriptions.

Table 2
Notations and their corresponding descriptions.

Notations	Description
G	causal graph
X	set of node variablesh
x_i	the ith node variable
$Pa(x_i)$	parent set of x_i
G_{x_i}	union of x_i and its ancestor set
Y	target variable
$f_{b_i}(\cdot)$	neural network function of the ith branch
$k(i)$	causal order of i
J_r	objective function for the regression task
J_c	objective function for the classification task

3.2. Preliminary concepts

In this paper, the causal relationships among variables are represented using a causal graph (Koller & Friedman, 2009). A causal graph is a directed acyclic graph, where each node x_i represents a variable, and the set of all variables is denoted as X. A directed edge $x_i \rightarrow x_j$ encodes a causal relationship, meaning that x_i is a direct cause of x_j or x_j is a direct effect of x_i. A causal graph is required to satisfy the causal edge assumption, which implies that all parent nodes are direct causes of their respective child nodes. For a given node x_i, the sets of its parent nodes, child nodes, and ancestor nodes are denoted as $Pa(x_i)$, $Ch(x_i)$, and $An(x_i)$, respectively. In addition to causal graphs, the underlying causal model can also be represented using a set of structural equations. A Structural Causal Model (SCM) (Pearl, 2009) consists of a set of endogenous variables V, a set of exogenous variables U, and a set of functions $\{f_i\}$, where each endogenous variable in x_i is generated as a function f_i of other variables. Every SCM implicitly defines or is associated with a corresponding causal graph.

A complete Structural Causal Model and its corresponding causal graph are often difficult to obtain in practice. However, partial and approximate causal information can be acquired through the following approaches: (1) **Domain-specific prior knowledge**, (2) **Causal discovery algorithms**, and (3) **A combination of prior knowledge and causal discovery algorithms**. Domain experts, researchers, or practitioners can iteratively construct the causal graph based on prior knowledge in their respective fields. However, the lack of a unified consensus may lead to inconsistencies in the resulting causal graph. Causal discovery algorithms alleviate the dependence on human input by recovering causal graphs directly from observational data, with their effectiveness largely determined by the algorithm's inherent capabilities. Ideally, the strengths of both prior knowledge and causal discovery algorithms should be integrated, leveraging data-driven insights alongside expert involvement to ensure the validity and credibility of the resulting causal graph. Based on such a causal graph, we can extract causal information related to the target variable, thereby mitigating the influence of spurious interactions among features on the target prediction.

3.3. The distribution of causal constraints

The learned distribution should approximate the distribution of the underlying causal model as closely as possible. Therefore, we eliminate cases that violate the conditional independence structures encoded by the causal graph. Specifically, we model the distribution of Y as a function of variables that are causally related to one another.

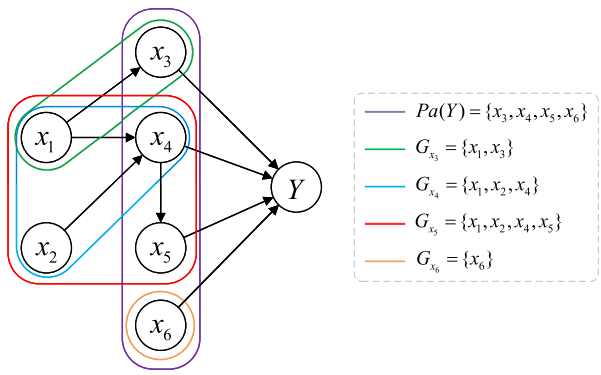


Fig. 2. A descriptive example of a causal grouping.

For any node $x_i \in Pa(Y)$, we define the following set of variables: $G_{x_i} := \{x_i\} \cup An\{x_i\}$. Then, we propose to restrict the learned model $f_Y(X)$ to the following class of functions:

$$f_Y(X) \sim f_Y(f(Pa(Y)), f(G_{x_1}), ..., f(G_{x_p})). \tag{1}$$

Eq. (1) indicates that by leveraging the information encoded in the causal graph, features can be grouped into distinct causal groups: (1) The parent set of the target variable Y, denoted as $Pa(Y)$, which directly influences the target prediction; (2) the ancestral sets G_{x_i}, consisting of any node $x_i \in Pa(Y)$, and its corresponding ancestor variables $An\{x_i\}$, representing indirect causal influences. This approach preserves all valid interactions with the target variable while discarding variables with spurious correlations. Specifically, we do not use the entire causal paths from the root nodes to the target variable as grouping criteria, as such paths may block certain valid interactions. A descriptive example of causal grouping is illustrated in Fig. 2. In this setting, the allowed interactions include $\{x_3, x_4, x_5, x_6\}$, $\{x_1, x_3\}$, $\{x_1, x_2, x_4\}$, $\{x_1, x_2, x_4, x_5\}$, and $\{x_6\}$. When the causal graph is not available, we can leverage domain knowledge about spurious interactions or apply causal discovery algorithms to define G_{x_i}. This approach helps constrain our model and mitigates its reliance on spurious interactions. The causal generative process of system variables consists of autonomous modules that do not influence each other, referred to as Independent Causal Mechanisms (ICM) (Schölkopf et al., 2021). Existing studies have explored the process of modeling all independent causal mechanisms to derive a Structural Causal Model. These investigations highlight both the challenges and advantages of such an approach, suggesting that it can be a suitable choice in specific scenarios. In this paper, causal groups contain all the information required for modeling the ICM of a given set of feature variables X for the target variable Y and its parent set $Pa(Y)$. Therefore, we aim to integrate neural network models with causal models. In the next section, we will elaborate on how to use a branch neural network to represent the structural information embedded in the causal model.

3.4. Problem definition

The formal problem definition studied in this paper is presented below.

Problem (causal prediction). Given an observational dataset $D = \{\{x_i^{(j)}\}_{i=1}^{p}, y^{(j)}\}_{j=1}^{n}$ and the corresponding variable set $X \cup \{Y\}$, our goal is to identify the parent variable set $Pa(Y)$ of the target variable Y, as well as the variable sets G_{x_i} composed of any node $x_i \in Pa(Y)$ and its ancestral variables $An\{x_i\}$. These groups are then fed into corresponding branch neural networks $f_{b_i}(\cdot)$, and their outputs are aggregated to produce a final estimation $\hat{y}^{(j)}$ that approximates the true value $y^{(j)}$ as closely as possible. Here, $x_i^{(j)}$ denotes the jth sample of the ith variable, and $y^{(j)}$ denotes the jth sample of the target variable Y. p and n represent the number of features and the number of samples, respectively. $f_{b_i}(\cdot)$ denotes the function of the ith branch neural network.

Please note that the key distinction between causal prediction and ordinary prediction lies in the type of input features used by the model. Ordinary prediction typically takes all observed features as inputs, whereas causal prediction relies on feature groups constructed according to an underlying causal structure. The former may introduce features that are spuriously correlated with the target variable, while the latter is able to effectively eliminate such spurious correlations. Consequently, causal grouping renders neural network models more interpretable and robust. Moreover, causal prediction is essentially supervised regression that utilizes causal grouping features, and it is not suitable for causal effect estimation. Causal effect estimation requires learning balanced representations (Johansson et al., 2016; Shalit et al., 2017), which is beyond the scope of this work.

4. The proposed framework

Inspired by prior theoretical analyses, we propose a novel framework called BNN-CC to address the problem (causal prediction) under study. This framework consists of three main components: (1) **Data Preprocessing**, (2) **Causal Grouping**, and (3) **Branch Neural Network**. The data preprocessing module handles tasks such as missing value imputation, data normalization, and dataset splitting. The causal grouping module aims to recover the underlying causal graph from observational data as accurately as possible.

Based on the discovered causal graph, we extract causal information related to the target variable Y, which includes the set of parent variables of Y, denoted as $Pa(Y)$, as well as the node sets G_{x_i} composed of any variable $x_i \in Pa(Y)$ and elements from its ancestor set $An\{x_i\}$. The branch neural network module consists of residual connections (He et al., 2016), attention mechanisms (Vaswani et al., 2017), and fully connected layers. The core challenge and contribution of this paper lie in how to effectively input the causal grouping information of the target variable into the branch neural network for training and inference. Specifically, the number of causal groups determines the number of branches in the network, while the number of features within each causal group defines the input dimension of the corresponding branch. First, each branch neural network incorporates a residual connection and the ELU activation function. Then, the outputs of all branches are mapped to a one-dimensional space and concatenated. An attention mechanism is applied to capture the most important feature information. Finally, a fully connected layer is used to make predictions on the target variable. The illustration of BNN-CC is shown in Fig. 3.

The following sections provide a detailed description of the Causal Grouping Module, the Branch Neural Network Module, and the Objective Function.

4.1. Causal grouping

We employ causal discovery algorithms to uncover the causal relationships among variables from observational data, thereby constructing a causal graph corresponding to the dataset. The Causal Grouping Module focuses on the target variable and extracts causal information consisting of its parent set and ancestor set. Based on the constructed causal graph, features are grouped according to their causal relevance to the target variable, assigning them to specific causal groups. The causal discovery algorithm adopted in this work is the LiNGAM, which was proposed by Shimizu et al. (2006). The LiNGAM assumes that the data generation process must satisfy the following three assumptions:

(1) **Directed Acyclic Graph**. The observed variables $\{x_i | i \in \{1, \dots, p\}\}$ are arranged in a specific causal order $k(i)$ such that cause variables always precede effect variables. In other words, the causal relationships among all variables can be represented by a directed acyclic graph (Pearl, 2009), where no feedback loops or cycles are allowed.
(2) **Linearity**. Each variable x_i is modeled as a linear combination of its corresponding causal variables, along with a noise term e_i and a

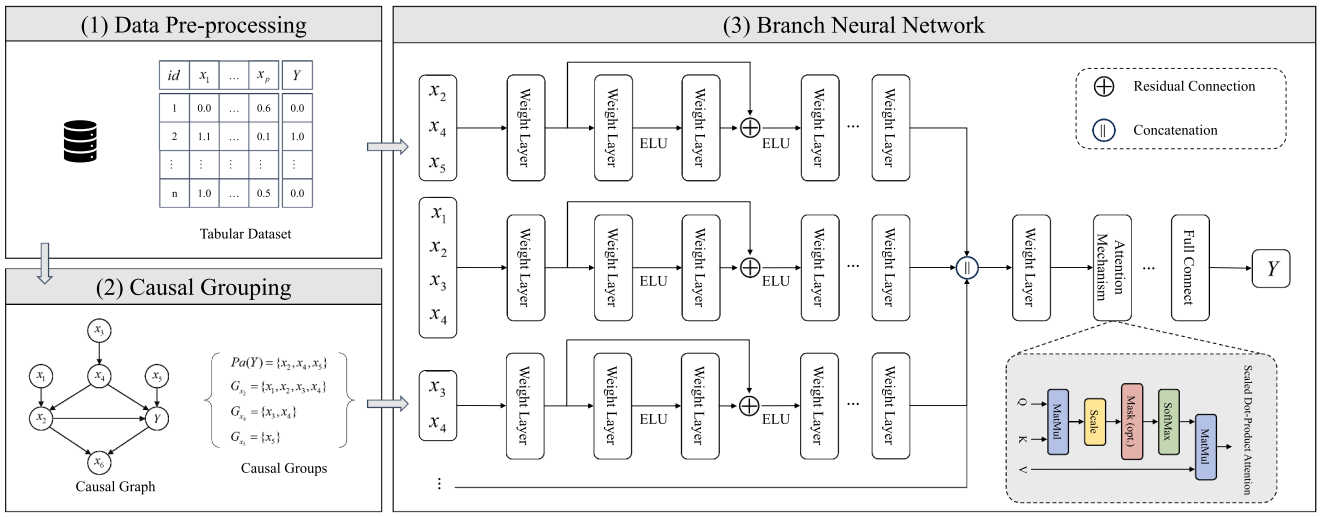


Fig. 3. An illustration of the proposed framework BNN-CC, which consists of three main components: (1) Data Pre-processing, (2) Causal Grouping, and (3) Branch Neural Network.

constant c_i. That is,

$$x_i = \sum_{k(j)<k(i)} b_{ij} x_j + e_i + c_i. \tag{2}$$

Here, $k(j) < k(i)$ indicates that the causal order[1] of x_j is before that of x_i, meaning that x_j is a cause of x_i. This causal relationship is denoted as $x_j \to x_i$.

(3) **Non-Gaussian and Independent Noise**. The noise variables e_i follow non-Gaussian distributions with non-zero variance and are mutually independent, i.e., $P(e_1, \dots e_p) = \prod_i P_i(e_i)$. Therefore, LiNGAM can be formally expressed as:

$$X = BX + E, \tag{3}$$

where $X \in \mathbb{R}^p$ is a random vector of observed variables, $B \in \mathbb{R}^{p\times p}$ is the weighted adjacency matrix representing causal relationships, $E \in \mathbb{R}^p$ is a vector of non-Gaussian, mutually independent noise variables. Due to the acyclicity assumption of the underlying causal graph, there exists a permutation matrix $P \in \mathbb{R}^{p\times p}$ such that the transformed matrix $B' = PBP^T$ is strictly lower triangular with zeros on the diagonal.

In this paper, the LiNGAM model is solved using the Independent Component Analysis (ICA) algorithm (Stone, 2004), i.e., ICA-LiNGAM. This algorithm first applies ICA to the observed data to obtain the linear transformation matrix W, where $S = WX$. Here, S is the vector of independent components, which corresponds to the noise variables in the structural causal model. By combining this linear transformation with Eq. (3), we observe a certain correspondence between matrices W and $(I - B)$, where I denotes the identity matrix. Given that B' is strictly lower triangular, the causal order of variables can be recovered from W through row and column permutations. Finally, a pruning procedure (such as the Wald test (Agresti, 2013) or Adaptive Lasso (Zou, 2006)) is employed to refine the learned structure and obtain the final causal graph. The specific steps of the ICA-LiNGAM algorithm are detailed in Algorithm 1.

Algorithm 1: ICA-LiNGAM.

Input: Given the dataset $X \in \mathbb{R}^{p\times n}$, where each row represents a variable vectorx_i, and n denotes the number of samples.

Output: The adjacency matrix B of the causal graph G.

1 Subtract the mean from each row of X, then apply the ICA algorithm to obtain a decomposition $X = AS$, where S has the same dimensions as X and its rows represent the independent components. Subsequently, we compute the inverse matrix $W = A^{-1}$.

2 Find a unique row permutation of the matrix W which yields a matrix $\tilde{W}$ without any zeros on its main diagonal, and hence the permutation is sought that minimizes $\sum 1/|\tilde{W}_{ii}|$.

3 Divide each row of $\tilde{W}$ by its corresponding diagonal element, to yield a new matrix $\tilde{W}'$ with all ones on the diagonal.

4 Compute the adjacency matrix $\hat{B} = I - \tilde{W}'$.

5 Determine the causal order K, i.e., find a permutation matrix which yields a matrix $\tilde{B} = P\hat{B}P^T$ which is a strictly lower triangular matrix.

6 Apply a pruning algorithm based on the determined causal order K to estimate the adjacency matrix B.

4.2. Branch neural network

The branch neural network consists of multilayer perceptrons,[2] residual connections, and an attention mechanism. Each branch maps its input to a one-dimensional space. The outputs from different branches are then concatenated, where the dimension of the concatenated vector depends on the number of branches. Finally, an attention mechanism is applied to the concatenated vector to highlight the relative importance of different causal groups. Branches assigned higher attention weights are regarded as making more significant contributions to the prediction of the target variable.

4.2.1. Residual connection

Formally, the residual block is defined as follows:

$$y = F(x, \{W_i\}) + x, \tag{4}$$

where x and y denote the input and output vectors (ordered arrays) of the considered layer, and W_i denotes the weight matrix of the ith hidden layer. The function $F(x, \{W_i\})$ represents the residual mapping to be learned. In this work, we adopt skip connections that span every two layers. For simplicity, bias terms are omitted in the formulation. The activation function used in $F = W_2\sigma(W_1 x)$ is the Exponential Linear Unit

[1] The cause precedes the effect; therefore, the time point at which the cause occurs is earlier than that of the effect.

[2] https://zh.d2l.ai/

(ELU) (Clevert et al., 2015), which is defined as:

$$\mathrm{ELU}(x) = \begin{cases} x, & \text{if} \quad x > 0 \\ \alpha \times (e^x - 1), & \text{if} \quad x \le 0 \end{cases}, \tag{5}$$

where α is a positive constant, with a default value of 1.0.

4.2.2. Attention mechanism

The attention function can be described as a mapping from a query and a set of key-value pairs to an output, where the query, keys, values, and output are all represented as vectors. The output is computed as a weighted sum of the values, with the weights determined by a compatibility function that measures the similarity between the query and each corresponding key. In this work, we adopt the scaled dot-product attention mechanism to compute attention scores, which is defined as follows:

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V. \tag{6}$$

where the inputs consist of the query $Q = W_q x \in \mathbb{R}^{n \times d_q}$, the keys $K = W_k x \in \mathbb{R}^{n \times d_k}$, and $V = W_v x \in \mathbb{R}^{n \times d_v}$. Here, W_q, W_k, and W_v are learnable projection matrices for the query, key, and value, respectively. The dot product between the query and all keys is first computed, scaled by the square root of the key dimension $\sqrt{d_k}$, and passed through a softmax function to obtain the attention weights. The softmax function is defined as:

$$\mathrm{softmax}(x_i) = \frac{e^{x_i}}{\sum_{j=1}^{p} e^{x_j}}, i = 1, \ldots, p. \tag{7}$$

where x_i and p is the ith element and dimension of x, respectively.

4.3. Objective function

Our proposed model is capable of handling both regression and classification tasks. Below, we present the objective function J (J_r for regression and J_c for classification).

$$J_r(x, y) = \sum_{i=1}^{n} (y^{(i)} - f_\theta(x^{(i)}))^2 + \lambda \|\theta\|_2^2. \tag{8}$$

$$\begin{aligned} J_c(x, y) = -\sum_{i=1}^{n} [y^{(i)} \log(f_\theta(x^{(i)})) + \\ (1 - y^{(i)}) \log(1 - f_\theta(x^{(i)}))] + \lambda \|\theta\|_2^2. \end{aligned} \tag{9}$$

For the regression task, the objective function consists of the standard mean squared error loss and a regularization (penalty) term. For the classification task, the objective function includes the cross-entropy loss and a regularization term L_2 as well. Let $x^{(i)}$ and $y^{(i)}$ denote the input feature vector and the true target value of the ith sample, respectively. Let $f_\theta(\cdot)$ represent the neural network with parameters θ, which is composed of branch neural networks $f_{b_i}(\cdot)$. The regularization term is controlled by a hyperparameter λ, and $\|\cdot\|_2^2$ denotes the squared L_2-norm of the model parameters.

To minimize Eq. (8) or (9), we adopt the Adam optimization algorithm (Kinga et al., 2015) to learn the optimal parameters of the model. Adam is an adaptive learning rate optimization method that combines the advantages of momentum and adaptive scaling. The parameter update rules of the Adam optimizer are defined as follows:

$$\theta_t = \theta_{t-1} - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon}. \tag{10}$$

Specifically, $\hat{m}_t = (\beta_1 m_{t-1} + (1 - \beta_1) g_t)/(1 - \beta_1^t)$; and $\hat{v}_t = (\beta_2 v_{t-1} + (1 - \beta_2) g_t^2)/(1 - \beta_2^t)$, where η denotes the learning rate (step size), typically set to 1×10^{-3}. ε is a small constant, such as 1×10^{-8} or 1×10^{-10}, introduced to avoid division by zero. The hyperparameters β_1 and β_2 are usually set to 0.9 or 0.999, respectively. β_1^t and β_2^t denote the tth power of β_1 and β_2. g_t represents the gradient at time step t. In summary, Algorithm 2 presents the pseudocode details of BNN-CC.

Algorithm 2: Branch neural networks based on causal constraints (BNN-CC).

Input: The training and testing set $(D_{tr}, D_{te}) \sim D$ with features X and a target variable Y.

Output: The network parameters θ of the BNN-CC model $f_\theta(\cdot)$.

```
G ← CD(D_tr) ;          // CD(·) and G represent a causal discovery algorithm and causal graph.
{G_{x_i}} ← G ; // Obtain the set of causal groups G_{x_i} from the causal graph G.
while i = 1, 2, ... do
    J ← f_θ({G_{x_i}}) ;     // Feed the feature data into the branch neural networks.
    update θ ← Adam{J} ;     // Update the parameters θ using Adam optimizer.
end
```

4.4. Discussion of model

This section revisits several implicit assumptions underlying the proposed model. First, we assume that the observational data reflects an underlying causal graph. Second, the causal graph is assumed to be free of unobserved confounders. Third, we assume that the causal groups of the target variable can be effectively represented by branch neural networks, and that their contribution to the prediction can be quantified via an attention mechanism. Despite these assumptions, the proposed framework demonstrates both generality and extensibility. In terms of generality, our model applies to both regression and classification tasks. In terms of extensibility, by configuring the BNN-CC model with multiple output heads, it can be adapted to support multi-task learning and counterfactual inference.

5. Experiments

We are particularly interested in whether models that incorporate causal information can outperform those that do not. To evaluate the effectiveness of our proposed BNN-CC model, we conduct extensive and diverse experiments on both synthetic datasets and two real-world datasets.

5.1. Baselines

We select classical and state-of-the-art machine learning methods as baselines for predicting the target variable. Different baselines are adopted for regression and classification tasks, respectively.

Baselines for Regression Tasks. The regression task aims to train and infer a continuous variable. The baseline methods we employ include Ordinary Least Squares Linear Regression (OLS) (Craven & Islam, 2011), k-Nearest Neighbors (k-NN) (Zhang, 2016), Decision Tree (DT) (Ying et al., 2015), Random Forest (RF) (Breiman, 2001), AdaBoost Regression (ABR) (Drucker, 1997), Epsilon-Support Vector Regression (SVR) (Smola & Schölkopf, 2004), Multi-Layer Perceptron (MLP) (Kruse et al., 2022), the practical version Invariant Risk Minimization (IRMv1) (Arjovsky et al., 2019) and a variant of the advanced method NN-SIP, namely NN-SIP-V (Pros & Vitrià, 2025).

Baselines for Classification Tasks. The classification task focuses on training and inferring a discrete variable, with this work specifically addressing binary classification. The baseline methods we employ include Logistic Regression (LR) (Best & Wolf, 2013), k-NN (Zhang, 2016), DT (Ying et al., 2015), RF (Breiman, 2001), AdaBoost Classification

(ABC) (Zhu et al., 2009), C-Support Vector Classification (SVC) (Smola & Schölkopf, 2004), MLP (Kruse et al., 2022), IRMv1 (Arjovsky et al., 2019) and NN-SIP-V (Pros & Vitrià, 2025).

Specifically, NN-SIP-V is a variant of the advanced NN-SIP network, adapted to predict either discrete or continuous target variables. Therefore, this variant is not designed for causal inference tasks but is instead suitable for the regression and classification tasks.

5.2. Evaluation metrics

The evaluation metrics used in this study for assessing performance in the regression task are introduced as follows. For regression tasks, we adopt three widely-used metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2) (James et al., 2023). The definitions of these metrics are provided below:

$$\mathrm{MAE}(y,\hat{y}) = \frac{1}{n}\sum_{i=1}^{n}|y^{(i)} - \hat{y}^{(i)}|, \tag{11}$$

$$\mathrm{RMSE}(y,\hat{y}) = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y^{(i)} - \hat{y}^{(i)}\right)^2}, \tag{12}$$

$$R^2(y,\hat{y}) = 1 - \frac{\sum_{i=1}^{n}(y^{(i)} - \hat{y}^{(i)})^2}{\sum_{i=1}^{n}(y^{(i)} - \frac{1}{n}\sum_{i=1}^{n} y^{(i)})^2}, \tag{13}$$

where $\hat{y}^{(i)}$ denotes the estimated (predicted) value of the ith sample, and $y^{(i)}$ is the true value. n represents the total number of samples. The values of MAE and RMSE range $[0,+\infty)$, where smaller values indicate that the predicted values are closer to the true values, and thus the regression performance of the model is better. The coefficient of determination R^2 range $[0,1]$, where values closer to 1 indicate superior regression performance.

For binary classification tasks, the two classes are typically referred to as positive and negative, with the target variable taking values 1 and 0, respectively. The evaluation metrics adopted in this study include Accuracy, F1-score, and the Area Under the Receiver Operating Characteristic Curve (AUC-ROC) (Fawcett, 2006). The definitions of Accuracy and F1-score are as follows:

$$\text{Accuracy} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{FP}+\mathrm{FN}+\mathrm{TN}}, \tag{14}$$

$$\text{F1-score} = \frac{2\times\mathrm{TP}}{2\times\mathrm{TP}+\mathrm{FP}+\mathrm{FN}}, \tag{15}$$

where TP (True Positive) and FP (False Positive) refer to the number of correctly predicted positive samples and incorrectly predicted positive samples, respectively; TN (True Negative) and FN (False Negative) refer to the number of correctly predicted negative samples and incorrectly predicted negative samples, respectively. The ROC curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at various threshold settings. The Area Under the ROC Curve (AUC) can be computed using the following formula:

$$\mathrm{AUC} = \frac{\sum_{i=1}^{n_1}\sum_{j=1}^{n_0} I(s_i > s_j)}{n_1 \times n_0}, \tag{16}$$

where $n_1 = \mathrm{TP}+\mathrm{FN}$ and $n_0 = \mathrm{TN}+\mathrm{FP}$ denote the number of true positive and negative samples, respectively. s_i represents the predicted score for the ith true positive sample, and s_j represents the predicted score for the jth true negative sample. The indicator function $I(\cdot)$ takes the value 1 if $s_i > s_j$, and 0 otherwise. The values of Accuracy, F1-score, and AUC all range from 0 to 1. The closer these values are to 1, the better the classification performance of the model. Additional evaluation metrics can also be considered to gain a more comprehensive understanding of model performance.[3] Below, we provide the hardware and software configuration details of the experimental environment, which can serve as a reference for reproducibility in future experiments.

- Hardware configuration: AMD Ryzen 7 5800H with Radeon Graphics (3.20 GHz) and 16GB of RAM.
- Software configuration: Python 3.8.19.

All observational data were standardized before being fed into the BNN-CC network for training and inference. We conducted experiments on six synthetic datasets and two public real-world datasets. The detailed hyperparameter settings used for training the BNN-CC model are summarized in Table 3. Specifically, the initial learning rate was set to 0.001 and decayed by a factor of 10 every 20 epochs. To efficiently and comprehensively recover the underlying causal graph from observational data, we adopt the ICA-LiNGAM or GES algorithm implemented by Zhang et al. (2021) to perform causal discovery. The data used for causal discovery are drawn from the training set of each dataset, with a typical training-test (training-validation) split ratio of 7:3. The output of the algorithm typically involves a subset of the feature variables, and our interest lies in the causal grouping of these features. The higher the quality of the discovered causal structure, the more reliable the resulting causal groups. Conversely, inaccurate causal discovery may lead to degraded model performance due to flawed grouping. Nevertheless, to leverage as much feature information as possible, we incorporate the remaining variables into a single causal group. ICA-LiNGAM is chosen for its robust convergence properties across diverse scenarios, and GES is applicable to the discovery of causal graphs for both continuous and discrete variables. It is worth noting that the choice of causal discovery algorithm is beyond the scope of this work.

Table 3
Hyperparameters for training the BNN-CC model on synthetic and real-world datasets.

Hyperparameters	Synthetic	IHDP	Twins
Initial Learning Rate (LeR)	0.001	0.001	0.001
LeR multiplicative factor	0.1	0.1	0.1
Epochs	300	300	100
Batch Size	15	15	5
Hidden Size	12	14	11
Hidden Layers	6	7	6
Dropout Rate	0.1	0.1	0.1
Residual Connection Size	2	2	2

5.3. Experiments on synthetic datasets

In general, it is challenging to obtain the underlying data-generating mechanisms and corresponding causal graphs from observational data, which makes it difficult to evaluate the outcomes of learned models. To address this problem, a common approach is to conduct experiments on synthetic datasets where both the data-generating process and the causal structures are explicitly defined. Therefore, this section presents a series of simulation results, which demonstrate the effectiveness and robustness of our proposed model.

5.3.1. Simulated graphs

Given a binary treatment variable $T \in \{0,1\}$, covariates $\{x_i | x_i \in x, i = 1,\dots,p\}$, and outcome variable Y, we construct a directed acyclic graph based on Fig. 2, where the data generating process implies the corresponding causal structure.

Construction of simulated graphs. To generate the synthetic datasets, we consider the number of samples $n \in \{200, 600\}$, the number of covariates $p \in \{20, 30, 40\}$, and the noise level $\sigma = 0.01$. The samples are evenly split into training and testing sets. Fig. 4 and Eq. (17) illustrate the corresponding causal graph and the data generating process (DGP), respectively.

[3] https://scikit-learn.org/stable/modules/model_evaluation.html

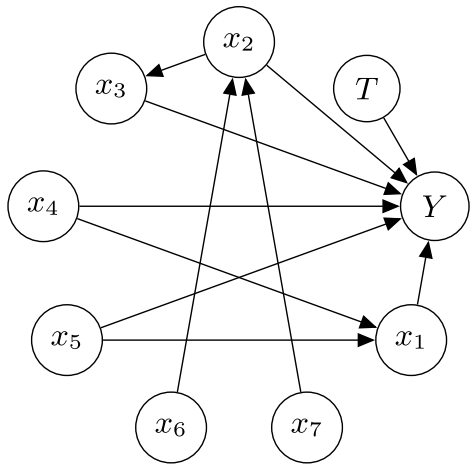


Fig. 4. The causal graph of synthetic dataset.

$$\begin{cases} x_1 = b_{11} \times x_4 + b_{12} \times x_5 + \varepsilon_1, \\ x_2 = b_{21} \times x_6 + b_{22} \times x_7 + \varepsilon_2, \\ x_3 = b_{31} \times x_2 + \varepsilon_3, \\ \tau = b_{41} \times x_1 + \log(1 + e^{x_2}), \\ \omega = \max(0, x_1 + x_2, x_3) + \max(0, x_4, x_5), \\ Y = (1 + (T - 0.5) \times \tau)^2 + e^{\omega} + \sigma \times \varepsilon_y. \end{cases} \tag{17}$$

A randomized trial is simulated in this study, where the input covariates $x_i \sim U(0,1)$ are generated from a multivariate distribution. The binary treatment variable T is generated by a fair coin flip, i.e., $T|x_i \sim Bernoulli(0.5)$; the noise variables $\varepsilon_1, \varepsilon_2, \varepsilon_y \sim U(0,1)$ and $\varepsilon_3 \sim U(0,2)$ are sampled from predefined distributions. The data generating process for the outcome variable Y is depicted in Eq. (17).

Fig. 4 illustrates that x_2 is a direct cause of x_3 and Y, indicating that it acts as a confounder. This introduces spurious correlations between x_3 and Y, which may hinder stable prediction. Therefore, it is necessary to perform causal grouping to enhance the interpretability and robustness of the model. For computational simplicity, we set the coefficients $b_{11} = b_{12} = b_{21} = b_{22} = b_{31} = b_{41} = 1$.

5.3.2. Results on synthetic datasets

Table 4 presents the experimental results on the synthetic datasets under different hyperparameter settings. The results demonstrate that under varying configurations of the data-generating process, specifically different sample sizes n and numbers of features p. Our model BNN-CC consistently outperforms all baseline methods as well as the state-of-the-art model NN-SIP-V (which shows the second-best performance). This performance gain is attributed to the incorporation of causal information in BNN-CC, which effectively mitigates the influence of irrelevant features on the prediction of the target variable. Consequently, our model significantly outperforms the MLP model that does not incorporate causal information. Therefore, incorporating causal information into prediction tasks not only enhances the interpretability and transparency of the model but also improves its predictive performance. Under a fixed noise level σ and sample size n, as the number of feature variables increases, the regression performance on the target variable is more likely to be affected by irrelevant variables. Nevertheless, the proposed BNN-CC model consistently outperforms the baseline methods.

To evaluate the quality of the causal graphs estimated by the algorithm, we employ graph-level metrics including Accuracy, Recall, F1-score, and Structural Hamming Distance (SHD) (Kalainathan et al., 2020). The SHD is a structural metric where lower values indicate better performance, while the remaining metrics yield better results with

Table 4
Experimental results under fixed noise level $\sigma = 0.01$ with different sample sizes n and numbers of feature variables p. Bold indicates the best performance, and underline indicates the second-best performance. The symbol "-" indicates that the model performs very poorly in fitting the target variable, where the R^2 value is negative.

Model	$(n,p) = (200,20)$			$(n,p) = (200,30)$			$(n,p) = (200,40)$			$(n,p) = (600,20)$			$(n,p) = (600,30)$			$(n,p) = (600,40)$		
	RMSE ↓	MAE ↓	R^2 ↑	RMSE ↓	MAE ↓	R^2 ↑	RMSE ↓	MAE ↓	R^2 ↑	RMSE ↓	MAE ↓	R^2 ↑	RMSE ↓	MAE ↓	R^2 ↑	RMSE ↓	MAE ↓	R^2 ↑
OLS (Craven & Islam, 2011)	0.537	0.380	0.707	0.603	0.444	0.630	0.524	0.392	0.720	0.538	0.394	0.709	–	–	–	–	–	–
k-NN (Zhang, 2016)	0.438	0.292	0.805	0.674	0.403	0.537	0.669	0.412	0.545	0.554	0.295	0.692	0.625	0.343	0.607	0.678	0.417	0.538
DT (Ying et al., 2015)	0.469	0.302	0.776	0.349	0.262	0.876	0.541	0.329	0.703	0.458	0.252	0.789	0.537	0.262	0.710	0.468	0.264	0.780
RF (Breiman, 2001)	0.406	0.222	0.833	0.349	0.205	0.876	0.378	0.234	0.855	0.357	0.190	0.872	0.431	0.217	0.813	0.421	0.250	0.822
ABR (Drucker, 1997)	0.407	0.226	0.832	0.321	0.225	0.895	0.398	0.269	0.839	0.359	0.272	0.870	0.446	0.304	0.800	0.500	0.348	0.749
SVR (Smola & Schölkopf, 2004)	0.391	0.234	0.844	0.573	0.308	0.667	0.593	0.379	0.643	0.405	0.210	0.835	0.493	0.201	0.756	0.556	0.293	0.689
MLP (Kruse et al., 2022)	0.474	0.368	0.771	0.656	0.485	0.563	0.644	0.475	0.578	0.330	0.208	0.891	0.454	0.271	0.793	0.508	0.359	0.741
IRMv1 (Arjovsky et al., 2019)	0.537	0.287	0.707	0.416	0.261	0.824	0.492	0.355	0.754	0.378	0.252	0.856	0.538	0.310	0.709	0.462	0.327	0.786
NN-SIP-V (Pros & Vitrià, 2025)	0.320	0.219	0.896	0.253	0.185	0.935	0.265	0.224	0.928	0.244	0.165	0.940	0.300	0.197	0.909	0.285	0.222	0.918
BNN-CC (Ours)	**0.298**	**0.171**	**0.910**	**0.201**	**0.120**	**0.959**	**0.233**	**0.187**	**0.945**	**0.163**	**0.093**	**0.973**	**0.291**	**0.115**	**0.915**	**0.272**	**0.199**	**0.925**

Table 5
Evaluation results of causal graph quality using ICA-LiNGAM under different sample sizes (n) and numbers of features (p).

(n, p)	Accuracy ↑	Recall ↑	F1-score ↑	SHD ↓
$(n = 200, p = 20)$	0.969	0.455	0.400	15
$(n = 200, p = 30)$	0.984	0.364	0.333	16
$(n = 200, p = 40)$	0.989	0.364	0.296	19
$(n = 600, p = 20)$	0.967	0.364	0.333	16
$(n = 600, p = 30)$	0.982	0.273	0.250	18
$(n = 600, p = 40)$	0.990	0.364	0.320	17
Average	0.980	0.364	0.322	16.830

higher values. Table 5 reports the graph-level metric results of ICA-LiNGAM across six different settings. The high accuracy indicates that the algorithm can effectively filter out features unrelated to the target variable, thereby preserving causal structural information about the target variable and providing a basis for subsequent causal grouping. The other graph-level metrics are relatively moderate, which is expected in a high-dimensional environment with limited samples. For example, under the hyperparameter setting $(\sigma, n, p) = (0.01, 200, 20)$, the discovered causal groups are $\{x_2, x_3\}$, $\{x_1, x_2, x_3, x_4, x_5\}$, $\{x_5\}$, and$\{x_3, x_4, x_5\}$. Based on Fig. 4, it can be observed that the causal groups obtained by the algorithm are consistent with the ground-truth causal structure besides $\{x_6, x_7, T\}$. Overall, these results provide strong evidence for the effectiveness and robustness of our model in causal modeling.

5.4. Results on real-world dataset

To further evaluate the effectiveness of the proposed framework, we conduct experiments on two public real-world datasets: the Infant Health and Development Program (IHDP) dataset and the Twins dataset.

5.4.1. Description of datasets

IHDP.[4] The dataset is composed of 747 instances with 25 feature variables (related to infants and their mothers, e.g., bw_x0, b_head_x1,…, was_x24), the treatment variable (home visit, home_visit_T), and target variable (cognitive test scores, cog_scores_Y).

Twins.[5] The dataset is composed of 23968 instances with 51 feature variables (related to parental characteristics, e.g., pldel_x0, birattnd_x1,…, wt_x50), the treatment variable (infant weight, weight01_T), and target variable (infant mortality, mortality_Y).

Table 6 presents selected descriptive statistics for the IHDP and Twins datasets, including the data type, mean, standard deviation (std), minimum value (min), 25th percentile (25%), 50th percentile (50%), 75th percentile (75%), and maximum value (max). For the IHDP and Twins datasets, the selected variables are those with a correlation intensity with the target variable of no less than 0.1 and 0.04, respectively. Please note that the meanings of variables in each dataset can be found on the GitHub repository.[6]

5.4.2. Correlation analysis on real-world datasets

To facilitate understanding of the relationships between variables in the datasets, we first conduct a correlation analysis (Rumsey, 2021). Fig. 5 illustrates the local correlation among variables in the datasets.

On the IHDP dataset, we calculate the pairwise correlation coefficients between selected variables (correlation strength greater than or equal to 0.1 with the target variable cog_scores_Y) to generate a local correlation heatmap, as shown in Fig. 5(a). On the Twins dataset, due to the presence of a large number of redundant features, we remove features whose absolute correlation coefficients with the target variable mortality_Y are less than 0.04, thereby reducing the feature dimension from 51 to 10. Subsequently, we calculate the pairwise correlation coefficients between the selected features and the target variable, resulting in the local correlation heatmap, as shown in Fig. 5(b). In both subfigures, blue elements indicate positive correlations, while red elements indicate negative correlations. The correlation coefficients within the red dashed box are sorted in descending order based on their correlation strength with the target variable. On the IHDP dataset, the top three variables in terms of correlation strength with the target variable cog_scores_Y are nnhealth_x4 (0.45), har_x20 (0.31), and home_visit_T (0.29), respectively. On the Twins dataset, the top three variables in terms of correlation strength with the target variable mortality_Y are wt_x50 (−0.62), gestat10_x15 (−0.31), and nprevistq_x40 (−0.16), respectively.

5.4.3. Causal groups on real-world datasets

We employ the causal discovery algorithms ICA-LiNGAM (Shimizu et al., 2006) and GES (Chickering, 2002) to infer the causal graphs corresponding to the IHDP and Twins datasets, respectively. Then we perform causal grouping of the features, as summarized in Table 7. These causal groups serve as critical inputs to the causal grouping module and are essential for data partitioning. Taking the Twins dataset as an example, the identified causal groups include {hydra_x22}, {incervix_x27}, and {hydra_x22, incervix_x27}, among others. These features are either direct parents or ancestors of the target variable mortality_Y. Each group is encoded and integrated by a branch neural network in our BNN-CC model. By leveraging this causal structure, the model conducts causal prediction of mortality_Y.

5.4.4. Results on real-world datasets

The experimental results on the two real-world datasets are presented below. Fig. 6 illustrates the loss and performance metric curves of our model BNN-CC on the validation set across training epochs. Specifically, Fig. 6(a) shows the loss (left axis) and the coefficient of determination R^2 (right axis) on the IHDP dataset, where the model begins to converge around the 50th epoch. Fig. 6(b) presents the loss (left axis) and accuracy (right axis) on the Twins dataset, where convergence starts around the 30th epoch. After multiple rounds of training, the BNN-CC model achieves the optimal performance on the IHDP and Twins datasets with $R^2 = 0.913$ and Accuracy = 0.916, respectively.

Table 8 presents the regression and classification performance results on the IHDP and Twins datasets (evaluated on the testing sets). For each dataset, the training and test sets are split using 10-fold cross-validation with a random seed of 42. Consequently, we do experiments with 10 replications on both the IHDP and Twins datasets, and report the mean and standard deviation (std) for each metric. The experimental results demonstrate that BNN-CC outperforms all baseline models (both regression and classification baselines), achieving state-of-the-art performance. On the IHDP dataset, the regression evaluation metrics are RMSE, MAE, and R^2. Our model BNN-CC achieves results (mean ± std) of RMSE = 0.312 ± 0.050, MAE = 0.231 ± 0.023, and $R^2 = 0.899 \pm 0.035$, respectively. The performance of NN-SIP-V ranks second with results of RMSE = 0.376 ± 0.101, MAE = 0.276 ± 0.047, and $R^2 = 0.847 \pm 0.091$. On the Twins dataset, classification evaluation metrics are Accuracy, F1-score, and AUC. Our model BNN-CC achieves the results (mean ± std) of Accuracy = 0.901 ± 0.007, F1-score = 0.710 ± 0.027, and AUC = 0.808 ± 0.021. The Accuracy of IRMv1 ranks second with the result of Accuracy = 0.901 ± 0.008. The F1-score and AUC of the LR model rank second with results of F1-score = 0.701 ± 0.025 and AUC = 0.799 ± 0.015.

Fig. 7 shows the critical different (CD) diagrams on the IHDP and Twins datasets. The average scores (R^2) on the IHDP dataset demonstrate that our model BNN-CC significantly outperforms the remaining baseline models. The average scores (F1-score) on the Twins dataset show that our model BNN-CC ranks first, though it exhibits no statistically significant difference compared to the LR model.

In summary, the proposed model demonstrates superior performance on both regression and classification tasks, outperforming all baseline

[4] https://www.fredjo.com
[5] https://www.nber.org/research/data/linked-birthinfant-death-cohort-data
[6] https://github.com/cjqshu/BNNCC-codes

Table 6
Descriptive statistics for the IHDP and Twins datasets. The selected variables are those whose correlation strength with the target variable is greater than or equal to 0.1 and 0.04, respectively.

IHDP Dataset									
ID	Variable	Data Type	Mean	Std	Min	25%	50%	75%	Max
#01	cog_scores_Y	Continuous	11.39	4.81	0.84	7.67	11.15	14.38	44.99
#02	nnhealth_x4	Continuous	0.00	1.00	−5.13	−0.57	0.12	0.68	2.37
#03	har_x20	Binary	0.16	0.36	0.00	0.00	0.00	0.00	1.00
#04	home_visit_T	Binary	0.19	0.39	0.00	0.00	0.00	0.00	1.00
#05	momage_x5	Continuous	0.00	1.00	−1.85	−0.86	−0.03	0.63	2.95
#06	tex_x23	Binary	0.13	0.33	0.00	0.00	0.00	0.00	1.00
#07	pen_x22	Binary	0.07	0.26	0.00	0.00	0.00	0.00	1.00
#08	b_marr_x8	Binary	0.52	0.50	0.00	0.00	1.00	1.00	1.00
#09	preterm_x2	Continuous	0.00	1.00	−1.85	−0.73	−0.36	0.76	2.99
#10	ein_x19	Binary	0.14	0.34	0.00	0.00	0.00	0.00	1.00
#11	mom_lths_x9	Binary	0.36	0.48	0.00	0.00	0.00	1.00	1.00
#12	was_x24	Binary	0.16	0.36	0.00	0.00	0.00	0.00	1.00
#13	bw_x0	Continuous	0.00	1.00	−2.73	−0.67	0.17	0.81	1.51
Twins Dataset									
#01	mortality_Y	Binary	0.18	0.38	0.00	0.00	0.00	0.00	1.00
#02	wt_x50	Continuous	1341	463	227	992	1440	1729	1999
#03	gestat10_x15	Ordinal	3.47	1.41	1.00	3.00	3.00	4.00	10.00
#04	nprevistq_x40	Ordinal	1.07	1.28	0.00	0.00	1.00	2.00	4.00
#05	hydra_x22	Binary	0.04	0.18	0.00	0.00	0.00	0.00	1.00
#06	csex_x16	Binary	0.49	0.50	0.00	0.00	0.00	1.00	1.00
#07	incervix_x27	Binary	0.01	0.11	0.00	0.00	0.00	0.00	1.00
#08	dmar_x8	Binary	0.63	0.48	0.00	0.00	1.00	1.00	1.00
#09	mpre5_x10	Ordinal	1.37	0.74	1.00	1.00	1.00	1.37	4.00
#10	phyper_x25	Binary	0.05	0.20	0.00	0.00	0.00	0.00	1.00
#11	mager8_x4	Ordinal	3.86	1.21	1.00	3.00	4.00	5.00	8.00

Table 7
Causal groups of the IHDP and Twins datasets.

IHDP Dataset	
ID	Causal Group
#01	{bw_x0}
#02	{bw_x0, b_head_x1}
#03	{preterm_x2, bw_x0}
#04	{mom_scoll_x11, mom_hs_x10, birth_o_x3, mom_lths_x9, first_x13}
#05	{preterm_x2, nnhealth_x4, bw_x0}
#06	{b_marr_x8, momage_x5, mom_scoll_x11, mom_hs_x10, mom_lths_x9}
#07	{b_marr_x8}
#08	{mom_scoll_x11, mom_hs_x10, mom_lths_x9, first_x13}
#09	{drugs_x15}
#10	{ark_x18}
#11	{ein_x19}
#12	{har_x20, mia_x21, ein_x19, tex_x23, ark_x18, was_x24, pen_x22}
#13	{mia_x21}
#14	{tex_x23}
#15	{home_visit_T}
#16	...
#17	{sex_x6, twin_x7, cig_x12, booze_x14, work_dur_x16, prenatal_x17}
Twins Dataset	
#01	{hydra_x22}
#02	{incervix_x27}
#03	{hydra_x22, incervix_x27}
#04	{ wt_x50, gestat10_x15, nprevistq_x40, csex_x16, dmar_x8, mpre5_x10, phyper_x25, mager8_x4, weight01_T}

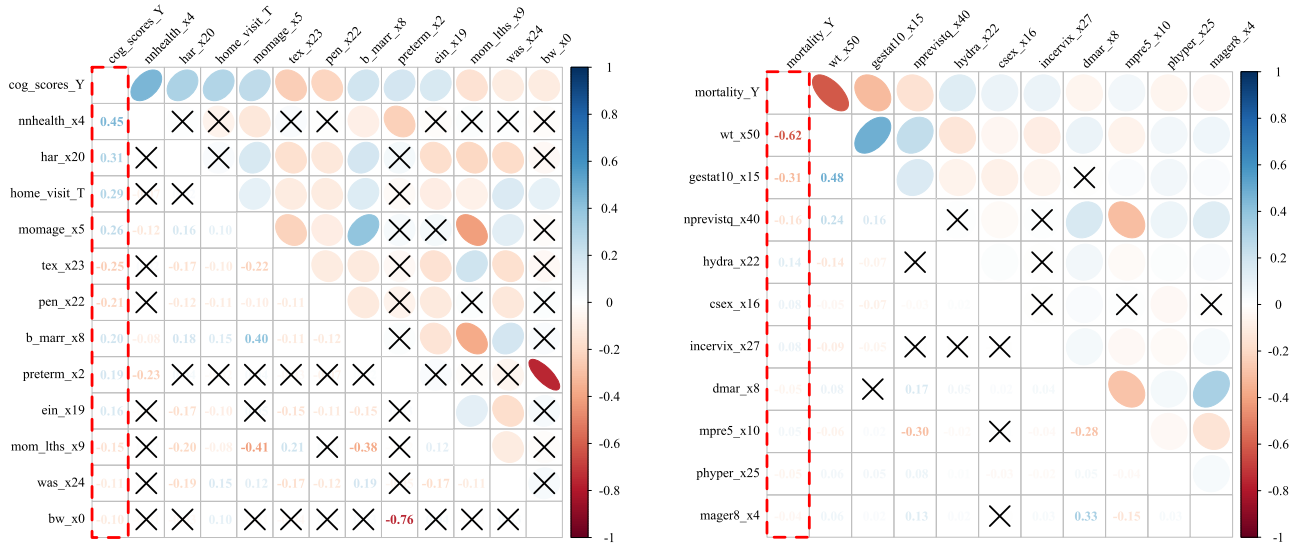
models. These results further validate the effectiveness and generalizability of BNN-CC.

5.5. Ablation study

To investigate the effectiveness of different components in our BNN-CC model, we conduct ablation studies by considering the following variant models:

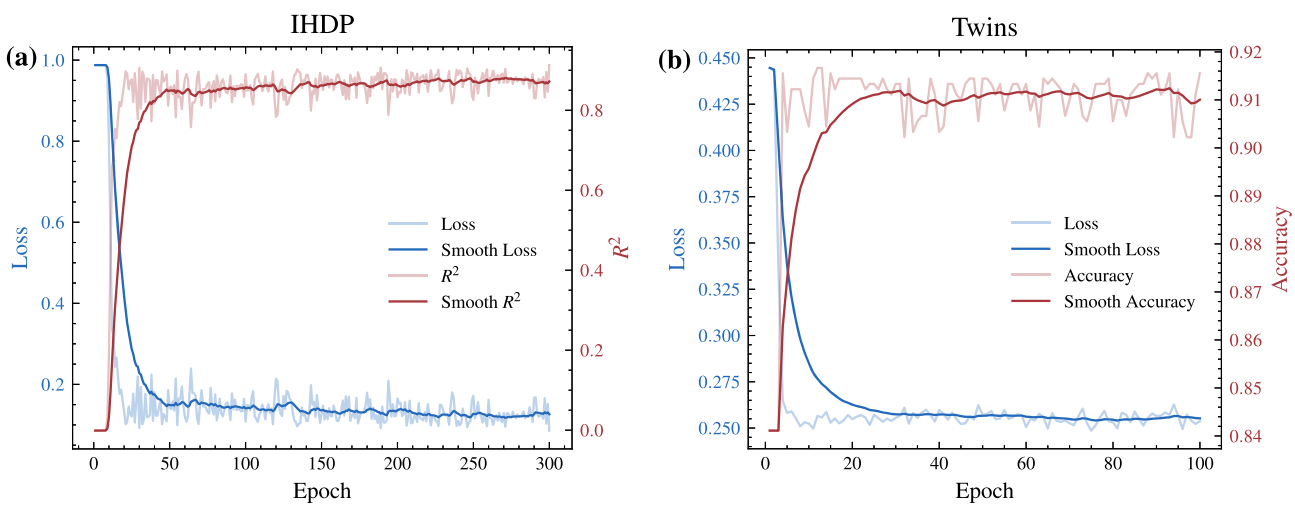
(1) **BNN-NCC**: A variant of BNN-CC without the causal grouping.
(2) **BNN-CC-NR**: A variant of BNN-CC without the residual connection.
(3) **BNN-CC-NA**: A variant of BNN-CC without the attention mechanism.

Table 9 present the ablation study results of the BNN-CC model on the IHDP and Twins datasets. On the IHDP dataset, we observe a significant performance gap between the variant model BNN-NCC and our proposed model BNN-CC. Specifically, the R^2 score of BNN-NCC drops by 0.122 compared to BNN-CC. This notable decline indicates that removing the causal grouping module leads to a substantial degradation in prediction performance, highlighting the effectiveness and importance of incorporating the causal grouping module in improving the model's predictive ability. We also observe that removing either the residual



(a) IHDP dataset. (b) Twins dataset.

Fig. 5. Heatmaps of pairwise variable correlations in the IHDP and Twins datasets. The correlation coefficients within the red dashed box are sorted in descending order based on the magnitude of their correlation with the target variable. The cross "×" in the figure indicates a correlation coefficient with no significant difference, i.e., mutually independent.

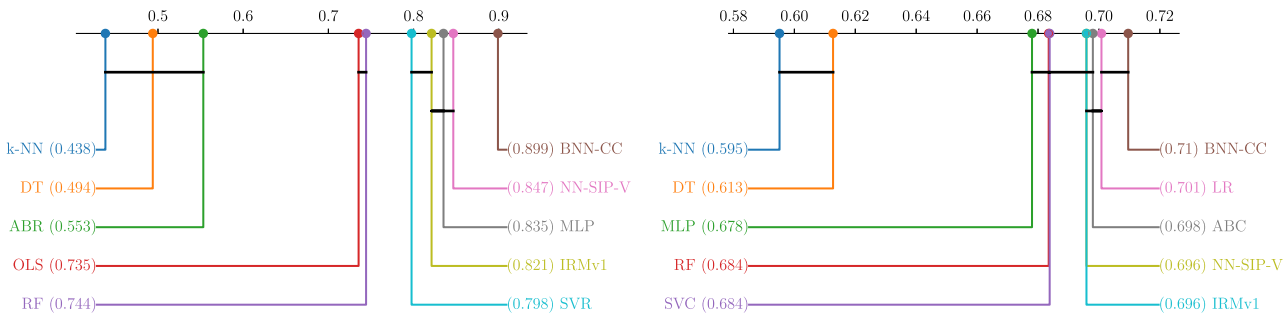


(a) Loss and R^2 on the IHDP validation set. (b) Loss and accuracy on the Twins validation set.

Fig. 6. Loss and performance metrics of the model on different datasets. The darker curves represent smoothed results using an exponential moving average.

Table 8
Regression and classification performance results (Mean ± Std) on two real-world datasets. Bold indicates the best result, and underlining denotes the second-best.

Model	IHDP Dataset			Twins Dataset		
	RMSE ↓	MAE ↓	R^2 ↑	Accuracy ↑	F1-score ↑	AUC ↑
OLS/LR (Best & Wolf, 2013; Craven & Islam, 2011)	0.510 ± 0.107	0.358 ± 0.042	0.735 ± 0.063	0.900 ± 0.010	0.701 ± 0.025	0.799 ± 0.015
k-NN (Zhang, 2016)	0.740 ± 0.100	0.565 ± 0.038	0.438 ± 0.088	0.872 ± 0.012	0.595 ± 0.043	0.734 ± 0.026
DT (Ying et al., 2015)	0.695 ± 0.100	0.489 ± 0.041	0.494 ± 0.135	0.867 ± 0.012	0.613 ± 0.045	0.754 ± 0.027
RF (Breiman, 2001)	0.502 ± 0.095	0.355 ± 0.046	0.744 ± 0.048	0.897 ± 0.009	0.684 ± 0.031	0.785 ± 0.019
ABR/ABC (Drucker, 1997; Zhu et al., 2009)	0.662 ± 0.114	0.516 ± 0.056	0.553 ± 0.078	0.899 ± 0.009	0.698 ± 0.028	0.798 ± 0.018
SVR/SVC (Smola & Schölkopf, 2004)	0.448 ± 0.102	0.319 ± 0.029	0.798 ± 0.045	0.899 ± 0.007	0.684 ± 0.022	0.783 ± 0.016
MLP (Kruse et al., 2022)	0.396 ± 0.071	0.300 ± 0.039	0.835 ± 0.051	0.895 ± 0.008	0.678 ± 0.042	0.784 ± 0.027
IRMv1 (Arjovsky et al., 2019)	0.415 ± 0.070	0.316 ± 0.043	0.821 ± 0.062	0.901 ± 0.008	0.696 ± 0.034	0.792 ± 0.025
NN-SIP-V (Pros & Vitrià, 2025)	0.376 ± 0.101	0.276 ± 0.047	0.847 ± 0.091	0.899 ± 0.007	0.696 ± 0.025	0.795 ± 0.015
BNN-CC (Ours)	**0.312 ± 0.050**	**0.231 ± 0.023**	**0.899 ± 0.035**	**0.901 ± 0.007**	**0.710 ± 0.027**	**0.808 ± 0.021**



(a) Average score (R^2) ranks on the IHDP dataset. (b) Average score (F1-score) ranks on the Twins dataset.

Fig. 7. Critical different (CD) diagrams on the IHDP and Twins datasets. The models closer to the right perform better. Models on the thick black line (crossbar) show no statistically significant difference, while those outside the thick black line exhibit statistically significant differences.

Table 9
Ablation study results on the IHDP and Twins datasets. The symbols ✗ and ✓ denote whether a component is removed or retained in the BNN-CC model, respectively.

Model	Causal Grouping	Residual Connection	Attention Mechanism	IHDP Dataset			Twins Dataset		
				RMSE ↓	MAE ↓	R^2 ↑	Accuracy ↑	F1-score ↑	AUC ↑
BNN-NCC	✗	✓	✓	0.454	0.350	0.791	0.912	0.708	0.815
BNN-CC-NR	✓	✗	✓	0.366	0.293	0.864	0.912	0.706	0.812
BNN-CC-NA	✓	✓	✗	0.326	0.249	0.892	0.913	0.702	0.804
BNN-CC	✓	✓	✓	**0.293**	**0.232**	**0.913**	**0.916**	**0.727**	**0.831**

connection or the attention mechanism results in a decline in predictive performance. This suggests that both components are indispensable to our model. The residual connection helps to mitigate issues such as vanishing and exploding gradients, while the attention mechanism amplifies the weights of representations derived from causal groups, thereby enhancing the model's overall predictive performance.

Fig. 8 presents the scatter plots of the true versus predicted values for the target variable on the IHDP dataset using BNN-CC and its variant models. Compared to the variant models, the predictions from BNN-CC lie closer to the black dashed line, indicating better agreement with the true values. This black dashed line is referred to as the ideal line or the line of equality (also known as the 1:1 line), which represents perfect prediction where the estimated values equal the true values. In addition, the Mean Absolute Percentage Error (MAPE) is defined as: $\mathrm{MAPE}(y^{(i)}, \hat{y}^{(i)}) = \frac{1}{n}\sum_{i=1}^{n}\left|(y^{(i)} - \hat{y}^{(i)})/y^{(i)}\right|$, where $y^{(i)}$ and $\hat{y}^{(i)}$ denote the true and predicted values. The value of MAPE lies within the range $[0, 1]$, and smaller values indicate that the predicted values are closer to the true values. Through the ablation study, we can evaluate the contribution of each component in the BNN-CC model to the prediction performance. The model BNN-CC, which retains all components, consistently outperforms its variant models, demonstrating the effectiveness of each component in enhancing model performance.

Similarly, on the Twins dataset, our proposed model BNN-CC outperforms the BNN-NCC variant across all metrics, including Accuracy, F1-score, and AUC.

5.6. Robustness verification

To validate the robustness of our BNN-CC model, we perform out-of-distribution (OOD) generalization experiments on the IHDP dataset. First, we create two distinct environments by selecting sex $X_6 \in \{0, 1\}$, where the training set consists exclusively of female ($X_6 = 0$) samples, whereas the testing set consists exclusively of male ($X_6 = 1$) samples. We then perform experiments using different grouping methods: Random-B randomly partitions three groups of features as inputs to the branch neural networks; CORR-B partitions three groups of features based on correlation coefficient magnitude as inputs to the branch neural networks; ANM-B, NOTEARS-B, and GOLEM-B respectively utilize causal grouping features of ANM, NOTEARS, and GOLEM (causal discovery methods) as inputs to the branch neural networks. Fig. 9 demonstrates that our model achieves optimal performance when environmental conditions change, yielding an R^2 (mean ± std) value of 0.879 ± 0.015 on the testing set. Compared to methods without causal grouping (Random-B and CORR-B), causal grouping methods (ANM-B, NOTEARS-B, and GOLEM-B) exhibit superior and more stable model performance.

5.7. Hyperparameter sensitivity analysis

To evaluate the robustness of the proposed BNN-CC model, we present in Fig. 10 its regression and classification performance under different model hyperparameter settings. More specifically, for the regression task on the IHDP dataset, we vary the values of key hyperparameters, including the batch size $\{5, 15, 20, 30\}$, the number of hidden layers $\{5, 6, 7, 8\}$, the hidden layer size $\{10, 12, 14, 16\}$, and the dropout rate $\{0.1, 0.2, 0.3, 0.4\}$. Through a grid search over hyperparameters, we identified the optimal configuration: a batch size of 15, 7 hidden layers, a hidden layer size of 14, and a dropout rate of 0.1. It can be observed that the regression performance is primarily influenced by the number of hidden layers and the hidden layer size, followed by the dropout rate and batch size.

Similarly, for the classification task on the Twins dataset, we vary the batch size $\{5, 10, 15, 20\}$, the number of hidden layers $\{6, 8, 10, 12\}$, the hidden layer size $\{5, 7, 9, 11\}$, and the dropout rate $\{0.1, 0.2, 0.3, 0.4\}$. The optimal combination of hyperparameters was found to be a batch size of 5, 6 hidden layers, a hidden layer size of 11, and a dropout rate of 0.1. For the classification task, our model BNN-CC demonstrates robustness across different settings of batch size, hidden layer size, and dropout rate. It can be observed that the classification performance is primarily influenced by the number of hidden layers.

In summary, for both regression and classification tasks, proper tuning of these hyperparameters is beneficial for achieving the optimal performance of the model BNN-CC.

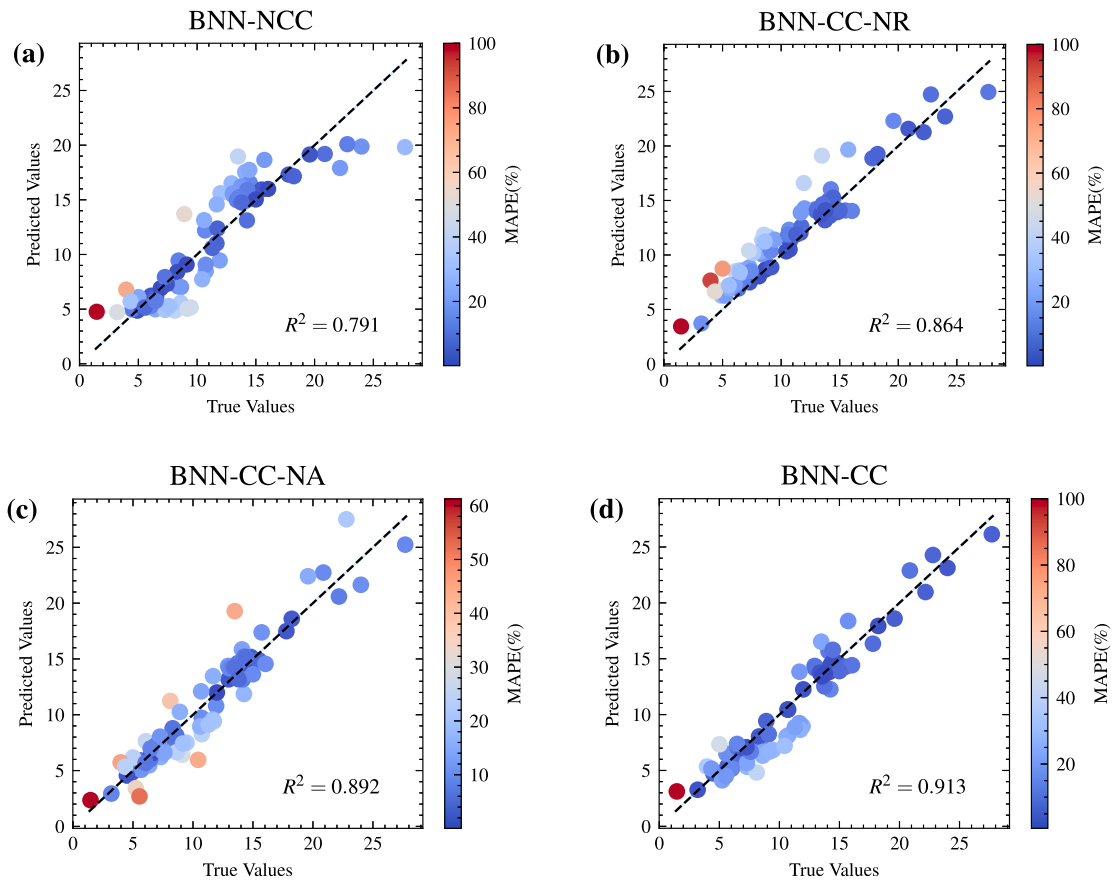
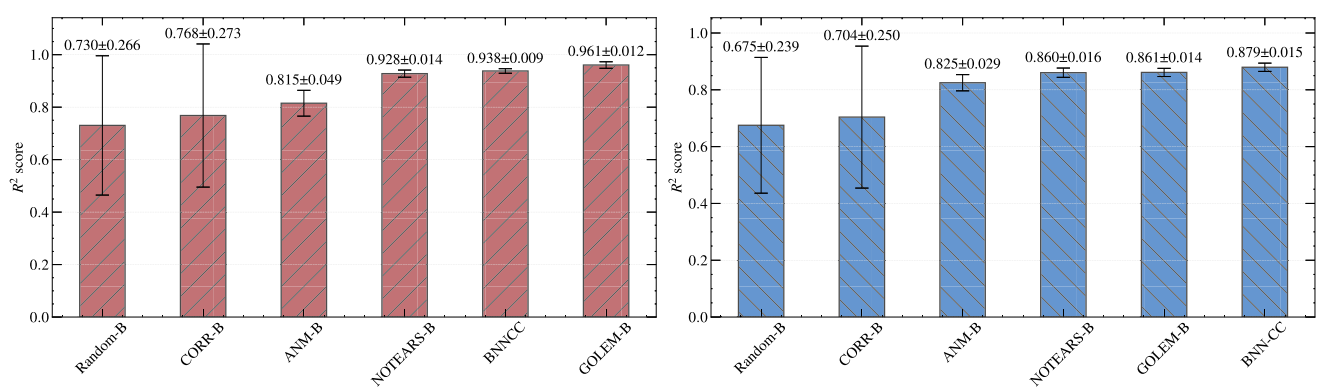


Fig. 8. Scatter plots of true vs predicted values of the target variable (cognitive test scores) on the IHDP dataset for BNN-CC and its variant models. The data points represent values in the original space (i.e., de-standardized values). MAPE denotes the mean absolute percentage error between the estimated and true values. A darker blue indicates a smaller error (i.e., predictions closer to the ground truth), whereas a darker red indicates a larger deviation.



(a) The IHDP training set R^2 comparison (Environment $X_6 = 0$). (b) The IHDP testing set R^2 comparison (Environment $X_6 = 1$).

Fig. 9. Out-of-distribution (OOD) generalization results on the IHDP dataset. The training and testing sets are drawn from different environments: the training set consists exclusively of female ($X_6 = 0$) samples, whereas the testing set consists exclusively of male ($X_6 = 1$) samples.

5.8. Discussion of experimental results

Our model BNN-CC outperforms both baseline and state-of-the-art models on both synthetic and real-world datasets. Under a predefined simulation graph, we constructed six synthetic datasets, and the experimental results consistently demonstrate that BNN-CC achieves the best performance, followed by the advanced model variant NN-SIP-V. On the real-world IHDP and Twins datasets, we first conducted both global and local correlation analyses to better understand the relationships among variables. We then apply a correlation threshold method to perform preliminary feature selection. After filtering, we extract the observational data of the selected features and the target variable. Then, we employ the causal discovery algorithms ICA-LiNGAM and GES to uncover the underlying causal graph implied by the observed data. From the resulting causal graph, we derive the causal groups related to the target variable. During the experiments, we observed that changing the index order of features within the causal groups alters the initialization of model parameters, which in turn leads to variations in model

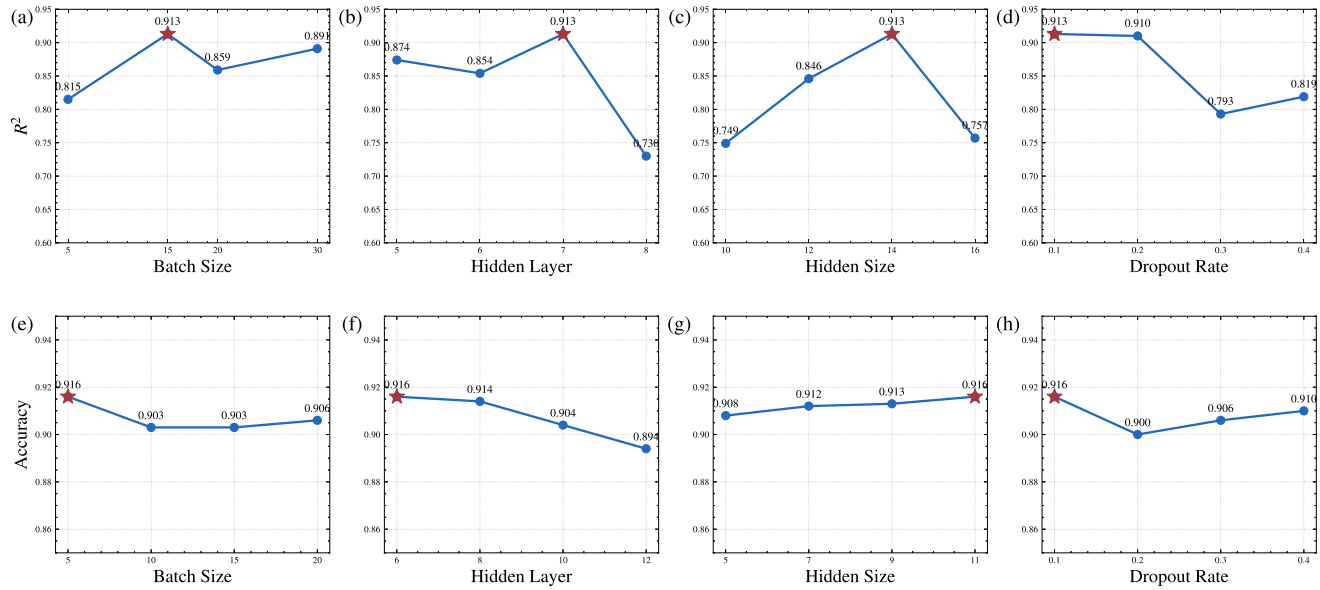


Fig. 10. Hyperparameter sensitivity analysis of the IHDP (a-d) and Twins (e-h) datasets. The considered hyperparameters consist of batch size, number of hidden layers, hidden size, and dropout rate.

performance. To ensure the reproducibility of experimental results, we fix both the order of feature indices within each causal group and the random seed during model input. The ablation study demonstrates that removing any individual component results in a performance drop. Notably, the removal of the causal grouping module leads to the most significant decline in performance, highlighting its critical role in our model. This is followed by the residual connections and the attention mechanism, both of which also contribute meaningfully to model effectiveness. OOD generalization experiments on the IHDP dataset show our model BNN-CC performance exhibits superior and more stable than baseline models. Finally, we conduct a sensitivity analysis on hyperparameters. The results indicate that batch size and dropout rate have relatively minor impacts on model performance, whereas the number of hidden layers and the size of each hidden layer have a more substantial influence. Moreover, we observe that the optimal combination of hyperparameters differs between regression and classification tasks.

6. Conclusion

In this paper, we investigate a critical problem in prediction using branched neural networks under causal constraints. To address the lack of causal interpretability in conventional neural networks, we propose a novel framework, BNN-CC, which leverages causally-constrained branched representation learning for target prediction. Specifically, we employ the causal discovery algorithms (ICA-LiNGAM or GES) to uncover the underlying causal graph from observational data. From the learned causal graph, we extract causal groups consisting of the parent or ancestor nodes of the target variable. Each group is then fed into a dedicated branch of the neural network, enabling the model to learn causally structured representations tailored for prediction. We conducted comprehensive experiments to evaluate the performance of the BNN-CC, and the results consistently demonstrate its effectiveness and superiority across various scenarios. Notably, the model achieved a coefficient of determination R^2 of 0.913 on the IHDP dataset and an accuracy of 0.916 on the Twins dataset. The effectiveness of each component is validated through ablation studies, with the causal grouping module demonstrating a particularly significant impact. To verify the robustness of our model, we conduct the OOD generalization experiments. Through sensitivity analysis of the model's hyperparameters, the optimal hyperparameter settings are determined. By incorporating causal information, the branched neural network model no longer relies solely on correlations among variables for prediction, but instead takes into account the underlying causal mechanisms in the observational data. This design makes BNN-CC more interpretable and robust, thereby offering reliable and valuable insights for decision-makers. The primary advantage of our proposed model lies in its incorporation of causal information to mitigate the issue of spurious correlations typically encountered in correlation-based neural network predictions. This causal integration enhances both the interpretability and robustness of the neural network. The causal grouping module disentangles feature interactions based on a discovered causal graph, enabling the formation of causally coherent feature groups. The branch neural network module then performs representation learning and aggregation for each causal group. This module includes both residual connections and an attention mechanism, which not only alleviates issues such as vanishing and exploding gradients but also empowers the model to focus on and capture the most informative features for the prediction task. One limitation of our model is that the derived causal groups are dependent on the quality of the observational data and the performance of the employed causal discovery algorithm. Extracting meaningful and credible causal graphs requires not only a large amount of data but also advanced causal discovery methods.

Future work will focus on improving the causal discovery algorithm and exploring more principled causal grouping strategies, such as grouping based on elements of the Markov blanket of the target variable. Additionally, the idea of causal grouping can be extended to various domains, such as causal inference and medical diagnosis. To further enhance the representational power of the model, more effective and expressive modules, such as Transformer-based architectures, can be integrated into the branch neural networks. Furthermore, the proposed model can be extended to multi-task learning, temporal tasks, and graph learning. For multi-task learning, our framework can be adapted by designing multiple output heads to predict multiple target variables simultaneously. For temporal tasks, dynamic causal graphs that evolve can be constructed, enabling causal grouping of temporally dependent features for improved forecasting performance. Lastly, in graph learning scenarios, causal grouping can be applied to node attributes, and the grouped features can then be fed into both branch neural networks and graph neural networks to perform causal prediction.

CRediT authorship contribution statement

Jiangqian Cai: Writing – original draft, Data curation, Software, Methodology, Investigation, Formal analysis, Visualization; **Quan Qian:** Conceptualization, Methodology, Funding acquisition, Project administration, Supervision, Writing – review & editing.

Data availability

The data and source code that support the findings are publicly available at https://github.com/cjqshu/BNNCC-codes.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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